

ADDIS ABABA UNIVERSITY

College of Technology and Built Environment (CTBE)

DATACENTER EXPERIENCE SHARING AND VISIT REPORT

AAiT Data Center (5-Kilo Campus) — Practical Observation Report

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Report Date	September 11, 2026 GC

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1. Introduction

1.1 Background

Modern organizations of all sizes — from commercial enterprises to academic institutions — depend critically on data center infrastructure to store, process, and transmit digital information. A data center is a dedicated physical facility that houses the IT systems, networking hardware, power distribution equipment, and environmental controls required to operate digital services continuously and reliably. As a discipline, data center design and management intersects computer networking, electrical engineering, structural engineering, and systems administration — making it a highly relevant subject for software engineering and information technology students.

Addis Ababa University's College of Technology and Built Environment (CTBE), formerly the Addis Ababa Institute of Technology (AAiT), operates its own data center at the 5-Kilo campus in Addis Ababa, Ethiopia. This facility serves as the backbone of the university's IT services, supporting academic operations, research, and administrative systems. It also functions as a hands-on teaching environment that brings classroom networking concepts to life for engineering and technology students.

1.2 Purpose and Objectives of the Visit

As part of the Fundamentals of Networking course (SITE-2102) at AAU CTBE, students were organized to visit and observe the operation of the AAiT Data Center. The visit was structured as an experience-sharing and guided tour session led by the data center's technical personnel. The objectives of this visit were:

- To observe how theoretical concepts in networking, power management, and systems administration are implemented in a real operational environment.
- To understand the physical layout, functional zones, and design principles of an institutional data center.
- To learn about the multi-layered power backup systems, cooling strategies, and fire suppression mechanisms employed.
- To understand the network topology, fiber infrastructure, and inter-campus connectivity of the AAiT data center.
- To gain practical insight into monitoring systems, modular scalability, and disaster recovery planning.
- To connect the academic knowledge gained in coursework with actual industry-standard practices.

1.3 Importance of Datacenter Infrastructure

Data centers are the foundation of the digital economy. For a university such as AAU — with over 50,000 students across 13 campuses, numerous digital portals (student portal, LMS platforms, admission systems, research repositories), and active research programs — reliable data center operation is not optional but essential. Downtime directly disrupts academic and administrative services, affects examinations, and interrupts student access to critical systems. Understanding how

such facilities are designed, protected, and managed is therefore fundamental knowledge for any graduate in software engineering or information technology.

2. Overview of the Datacenter Facility

The AAiT Data Center — now operating under the College of Technology and Built Environment (CTBE) — is located on the ground floor of the NB Building on the 5-Kilo campus of Addis Ababa University. The choice of the ground floor is deliberate: data center equipment is exceptionally heavy, and upper floors require additional structural reinforcement to accommodate the load of servers, UPS battery banks, and power equipment. The ground floor placement eliminates structural risk and simplifies the logistics of equipment installation and replacement.

The facility spans approximately 300.9 square meters in total, divided into three primary functional zones: a Power Room (approximately 28.8 m²), a Server/IT Room (approximately 54.81 m²), and a Network Operations Center (NOC) room (approximately 14.4 m²). This layout mirrors international best practice, which separates power infrastructure from IT equipment and centralizes monitoring in a dedicated operations room.

The visit session was conducted as a guided tour and knowledge-sharing session led by the data center's technical staff. Students were walked through each zone of the facility and given detailed explanations of the function, design rationale, and operational characteristics of each system encountered. The session was interactive, with staff answering technical questions from students throughout the tour.

Zone	Approximate Area	Primary Function
Power Room	~28.8 m ²	AVR, ATS, UPS systems, battery banks, and power distribution
Server / IT Room	~54.81 m ²	Rack-mounted servers, storage, networking equipment, and hot/cold aisle containment
NOC Room	~14.4 m ²	Centralized monitoring and management of all data center systems

Table 1: Functional zones of the AAiT Data Center

3. Technical Infrastructure and Observations

3.1 Server and Computing Infrastructure

The server room houses rack-mounted servers that provide the full range of digital services operated by the university. The data center staff described the servers as high-capacity machines configured to deliver different functional roles across the university network, including web hosting, email services, media services, file storage, student portals, learning management systems, and big data processing.

The facility uses 42U standard racks — meaning each rack can accommodate up to 42 rack units of equipment. At the time of the visit, approximately 6–8 racks are installed and operational, with the facility currently utilizing approximately 30% of its total rack capacity. This deliberate under-utilization reflects a planned approach to scalability: the data center was designed with future growth in mind, and capacity was provisioned well beyond immediate needs.

For research and computational workloads, the data center hosts a high-performance computing (HPC) cluster featuring 8 NVIDIA GPUs and 1.6 TB of total RAM, used primarily for AI/ML model training, scientific simulations, and research analysis by faculty and postgraduate students.

Component	Specification / Observation
Rack type	Standard 42U server racks
Installed racks	6–8 racks operational
Maximum rack capacity	Up to 48 racks (Huawei FusionModule 2000 system ceiling)
Current utilization	~30% of total rack capacity
Server roles	Web, mail, media, file, portal, LMS, big data, simulation
HPC resources	8 NVIDIA GPUs, 1.6 TB RAM
Rack height standard	42U per rack

Table 2: Server and computing infrastructure summary

3.2 Networking Infrastructure

The networking infrastructure of the AAiT data center is built on a fiber optic backbone that connects every building on the campus. Each building connection terminates at a dedicated fiber patch panel within the data center, making it the central aggregation point for all campus network traffic.

3.2.1 Fiber Type Selection

The data center staff explained that the entire campus fiber infrastructure uses single-mode fiber optic cable. The choice between single-mode and multi-mode fiber depends on distance: multi-mode fiber (with sub-variants OM1, OM2, OM3) is suitable for short distances, while single-mode fiber supports much longer distances with lower signal attenuation. Since inter-building distances across the AAiT campus exceed 400–500 meters, single-mode fiber was selected as the appropriate standard for all connections.

3.2.2 Network Topology — Collapsed Core

The campus network employs a collapsed core architecture. In a traditional three-tier network design, access, distribution, and core switches form separate layers. In a collapsed core design, the distribution layer is eliminated and access switches connect directly to the core switches. This approach reduces the number of network hops, simplifies management, lowers cost by eliminating distribution-layer hardware, and is well-suited for medium-scale campus environments such as AAiT. Each access switch from across the campus connects directly to one of the two core switches housed in the data center server room.

The data center staff noted that for very large or very high-traffic environments, a leaf-and-spine architecture would be more appropriate, but the collapsed core model is sufficient and cost-effective for the current scale of operations.

3.2.3 Redundancy and Link Resilience

To ensure continuous network availability, the data center implements redundancy at multiple network layers:

- Two core switches: If one core switch fails, the other takes over seamlessly.
- Two firewalls (Fortinet): Dual Fortinet firewalls perform stateful packet inspection of all incoming and outgoing traffic, providing both security and redundancy.
- Dual fiber paths per access switch: Each access switch receives two fiber uplinks — one routed through the parking area and one routed through a secondary building path. If one physical fiber path is cut or fails, the other link automatically takes over.
- Multiple campus inter-links via Ethio Telecom: The data center maintains 3–4 inter-campus links (to 4 Kilo campus, 6 Kilo campus, Dental campus, and others) through Ethio Telecom wavelength services. If one inter-campus path fails, traffic is rerouted through an alternative path.

A practical example of this inter-campus redundancy was shared during the visit: when the 4 Kilo campus experienced a cabinet network issue, its internet traffic was automatically rerouted through the AAiT 5-Kilo data center, allowing 4 Kilo users to continue accessing online resources without disruption.

Network Component	Specification / Implementation
Fiber type	Single-mode fiber (campus-wide, distances >400m)
Topology	Collapsed core (access switches → core switches directly)
Core switches	2 (redundant pair)
Firewalls	Fortinet (2 units — redundant, stateful packet inspection)
Link redundancy	Dual fiber paths per access switch (separate physical routes)
External connectivity	Ethio Telecom fiber wavelength service
Inter-campus links	3–4 links (4 Kilo, 6 Kilo, Dental, and others)
Patch panels	Fiber patch panels per building connection

Table 3: Networking infrastructure summary

3.3 Storage and Data Management

Data generated and consumed by university services is stored on the servers within the AAiT data center. Backup copies of critical data are replicated to the AAU 6-Kilo campus data center on a scheduled basis. The frequency of backup operations depends on the criticality of the data: financial and mission-critical systems are backed up daily, while less critical systems follow a less frequent schedule determined by their business requirements.

The data center staff were clear that the 5-Kilo and 6-Kilo campus data centers currently serve a mutual backup role for each other — each site acts as a warm standby for the other. However, staff acknowledged that this arrangement does not fully satisfy the requirements of a formal Disaster Recovery (DR) site, which should be separated by at least 100 kilometers from the primary site to ensure that a single regional event (such as a natural disaster, power grid failure, or fire) cannot affect both sites simultaneously.

3.4 Power and Backup Systems

Power infrastructure is one of the most critical and carefully designed elements of any data center. The AAiT data center employs a multi-layered power architecture designed to eliminate single points of failure and ensure that IT equipment receives clean, stable, and uninterrupted power under all conditions.

3.4.1 Utility Power Supply

Commercial utility power is supplied by the Ethiopian Electric Utility (EEU), historically known as EELPA. The data center staff explained that for environments with a lower power criticality requirement, a single utility feed is sufficient; however, for highly critical environments (such as hospitals or financial institutions), utility power is sourced from two independent directions (e.g., north and south distribution paths) so that a fault on one utility feed does not interrupt power delivery. At the AAiT data center, the current arrangement uses a single utility feed, supplemented by robust backup systems.

3.4.2 Automatic Voltage Regulator (AVR)

Incoming utility power first passes through an Automatic Voltage Regulator (AVR). The AVR continuously stabilizes the incoming voltage, compensating for fluctuations and voltage sags that are common in the Ethiopian power grid. This protects sensitive IT equipment from voltage stress and ensures that downstream systems receive stable power.

3.4.3 Automatic Transfer Switch (ATS)

From the AVR, power flows into an Automatic Transfer Switch (ATS). The ATS continuously monitors the utility power supply. When a power failure is detected, it automatically initiates the generator startup sequence without any manual intervention. Once the generator has stabilized and is supplying power, the ATS transfers the load to generator power. When utility power is restored, the ATS performs the reverse transfer — switching the load back to utility power and shutting down the generator — again automatically.

3.4.4 UPS Systems and Battery Banks

The Uninterruptible Power Supply (UPS) system — comprising multiple UPS 5000E series units — bridges the transition gap between utility power failure and generator startup. When utility power is lost, the UPS immediately supplies power from its battery banks without any interruption to connected equipment. The ATS then signals the generator to start, and after approximately one minute, the generator reaches stable operating speed and takes over the load from the UPS. The battery banks are sized to provide approximately 8 hours of runtime, ensuring that even extended generator failures can be managed without data loss or equipment damage. Additional battery modules can be installed on either side of the UPS rack to increase runtime capacity as business demands grow.

3.4.5 Backup Generators

The data center is served by two independent generators: a dedicated mini-generator specifically for the data center load, and a larger campus-wide generator serving broader university infrastructure. This dual-generator arrangement provides an additional layer of protection: if the primary generator fails to start within the ATS's waiting period (~1 minute), the secondary generator is available as a fallback, while the UPS continues to support the load throughout the transition.

3.4.6 Power Capacity

The current power capacity of the data center is 80 kVA. This is contrasted with the Huawei FusionModule 2000 standard, under which each individual rack module can support up to 140 kVA. The data center staff emphasized that power infrastructure must scale in proportion with IT infrastructure: it is not possible to add racks without simultaneously scaling up power capacity and generator sizing.

Power Component	Detail
Utility supply	Ethiopian Electric Utility (EEU / EELPA)
AVR	Automatic Voltage Regulator — stabilizes incoming voltage
ATS	Automatic Transfer Switch — switches to generator on power failure
UPS model	UPS 5000E series (multiple units with visual displays)
Battery runtime	~8 hours (expandable by adding battery modules)
Generator 1	Dedicated data center mini-generator
Generator 2	Campus-wide generator (secondary backup)
Generator start delay	~1 minute (UPS covers this gap)
Current power capacity	80 kVA
Huawei module max (future)	140 kVA per rack module

Table 4: Power and backup systems summary

Power Flow Architecture

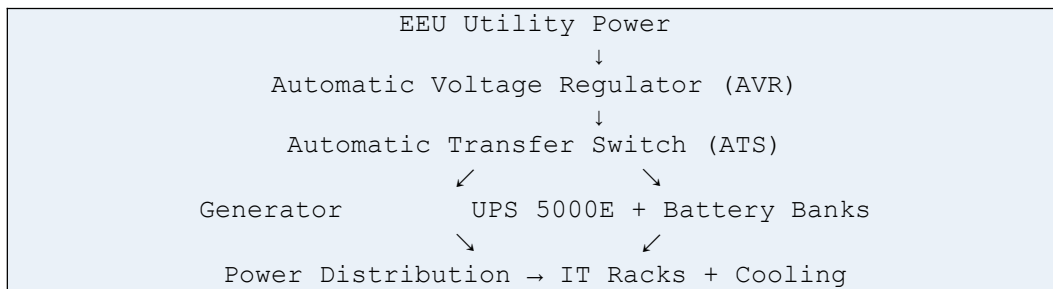


Figure 1: Power flow architecture of the AAIT Data Center

3.5 Cooling and Environmental Control

Heat management is described by the data center staff as one of the most significant ongoing challenges in data center operations. Every electronic device that consumes power generates heat as a byproduct; as the density of equipment increases, so does the thermal load. Without effective cooling, equipment temperatures rise rapidly, leading to performance degradation, accelerated component aging, and ultimately hardware failure.

3.5.1 Hot Aisle / Cold Aisle Containment

The server room implements a hot aisle / cold aisle containment strategy. Server racks are arranged back-to-back in alternating pairs: the front faces of servers face the cold aisles (where cool air is delivered), and the rear exhaust faces the hot aisles (where expelled hot air is captured). This separation prevents the mixing of cool intake air with hot exhaust, dramatically improving cooling efficiency.

Cool air is drawn in through inlets at the base of the cold aisle. Hot exhaust air from server rears is captured in the hot aisle and routed through ducts to the outdoor AC units, which extract the heat and expel it outside the building. The outdoor AC units are visible on the building exterior.

3.5.2 Cooling Units

The power room is cooled by cassette-type indoor air conditioning units — ceiling-mounted units that distribute conditioned air evenly throughout the room. Multiple types of AC units exist in data center environments (floor-standing units, wall-mount units, and cassette ceiling units), and the appropriate type is selected based on the room geometry and airflow requirements.

3.5.3 Environmental Monitoring

Temperature and humidity sensors are deployed throughout the data center — on the floor, at rack level, and above the containment structure. All sensor readings are aggregated by the ECC 800 device and visualized in the NetEco monitoring application, which displays real-time environmental status with severity levels (normal, moderate, critical). This allows operations staff to detect and respond to thermal anomalies before they cause equipment damage.

3.6 Physical Security and Access Control

Physical access to the data center is controlled through electronic access control systems on all entry doors, including the power room, server room, and NOC. These systems are integrated with the overall monitoring infrastructure — door status and access events are visible from the NOC and the NetEco platform. CCTV surveillance covers the facility as part of the broader AAU Smart Campus infrastructure deployed through the Ethio Telecom partnership.

A key operational practice described by the data center staff is the use of a staging process for new equipment. Any device brought into the facility must first be isolated and evaluated — typically in the NOC room, which serves as an informal staging area — before being connected to the production network. This prevents rogue or compromised devices from being directly connected to the data center infrastructure, reducing the risk of security incidents.

Faculty and departmental users are allocated isolated resources through per-faculty VLAN segmentation. Each faculty or department receives its own VLAN, dedicated VM space, port assignments, and — where required — separate routing configurations and system administrators. This provides logical separation of data and systems even when sharing the same physical infrastructure.

3.7 Fire Protection and Safety

The data center employs a gaseous fire suppression system using FM-200 (HFC-227ea), a clean-agent suppression gas that is highly effective at extinguishing fires without damaging electronic equipment or leaving residue. FM-200 is widely used in server room environments as an alternative to water-based suppression, which would cause catastrophic damage to IT equipment.

FM-200 cylinders are installed to serve both the server room and the power room. The fire detection and suppression system is integrated with the access control and monitoring infrastructure: upon detecting smoke or heat, the system automatically triggers an alert, and if the event escalates, it autonomously activates the gas suppression system — even when no personnel are present. Fire exits are present above and below the data center as per standard safety requirements.

The data center staff also highlighted that the wall cladding materials used in the facility are fire-rated and thermally efficient. These specialized clads are capable of absorbing and retaining cold air while also resisting heat penetration for an extended period, providing passive fire resistance in addition to the active suppression system. Class B and Class C rated bricks and materials are used in the wall construction, consistent with data center construction standards.

3.8 Monitoring and Management Systems

All data center systems — UPS units, AC cooling units, access control devices, environmental sensors, and network devices — are connected to a central aggregation device known as the ECC 800. The ECC 800 collects operational telemetry from all connected systems and communicates this data to the NetEco monitoring application developed by Huawei. NetEco provides a unified dashboard for

operations staff, displaying the status of power, cooling, environmental conditions, and device health across the entire facility.

Each UPS unit also has its own integrated visual display panel, providing local status information (input source, output status, battery level, error conditions) directly on the device itself. This is important for physical troubleshooting when staff are working in the power room.

The data center staff also noted the availability of open-source monitoring alternatives such as Nagios, which can be used when vendor-specific integration is not required. The choice between vendor-specific tools (such as NetEco) and open-source platforms involves tradeoffs between native device integration quality and licensing cost.

System	Tool / Platform	Function
Data aggregation hub	ECC 800	Collects telemetry from all devices (UPS, AC, sensors, access control)
Monitoring platform	Huawei NetEco	Unified dashboard — power, cooling, environment, device status
UPS local display	UPS 5000E screen	Local status panel on each UPS unit
Open-source option	Nagios	Alternative monitoring for non-vendor-specific use cases
NOC room	Dedicated operations hub	Real-time monitoring and management center for all systems

Table 5: Monitoring and management systems

3.9 Reliability, Redundancy, and Availability

A recurring theme throughout the visit was the concept of redundancy — designing every critical system such that a single failure does not cause a service outage. The data center staff summarized the redundancy approach as having multiple units of every critical component, with automatic failover to the backup unit if the primary fails:

System Layer	Redundancy Implementation
Utility power	Single feed (dual-feed design principle discussed for critical environments)
Voltage regulation	AVR stabilizes incoming power continuously
Generator backup	2 generators (data center mini-generator + campus-wide generator)
UPS bridging	UPS 5000E with ~8 hours battery autonomy
Core switches	2 core switches (automatic failover)
Firewalls	2 Fortinet firewalls (redundant pair)
Fiber paths	Dual fiber uplinks per access switch (separate physical routes)
Inter-campus links	3–4 Ethio Telecom paths to multiple campuses
Cooling units	Multiple cassette AC units (power room) + outdoor ACs (server room)
Data backup	5-Kilo ↔ 6-Kilo campus mutual backup (scheduled replication)

Table 6: Redundancy implementation across system layers

4. Experience-Sharing Session — Key Professional Insights

Beyond the physical tour, the data center personnel conducted an informative knowledge-sharing session, sharing professional insights that extended beyond what is typically covered in textbooks. The following represent the most significant points discussed:

4.1 Business-Driven Data Center Design

The staff emphasized repeatedly that data center design must always begin with the business requirements of the organization it will serve. The degree of power redundancy, the number of generators, the sizing of UPS systems, and the level of fire protection should all be calibrated to the criticality of the services being hosted. A hospital or financial institution requires a much higher level of investment in redundancy than an academic institution.

4.2 The Huawei FusionModule 2000 — Modular Scalability

The data center uses the Huawei FusionModule 2000 containment and infrastructure system. One of its key advantages highlighted by staff is modular scalability: the system can be expanded by adding modules incrementally, without requiring a full infrastructure rebuild. Each module can support racks with power densities of up to 140 kW per rack. The full system can scale to support up to 48 IT racks in total. Staff demonstrated how expansion works: a new containment module is attached to the existing infrastructure and new racks, cooling units, and power connections are added in parallel.

The staff also explained the rationale for selecting a single-vendor integrated solution over a multi-vendor approach: in a multi-vendor environment, the temperature controller, servers, and cooling systems come from different companies, requiring different specialists, different management tools, and creating integration challenges when systems need to communicate with each other. A fully integrated single-vendor solution like the Huawei FusionModule 2000 with NetEco management simplifies operations, reduces the number of specialists required, and provides a unified management interface.

4.3 Structural and Site Considerations

The staff discussed the importance of structural engineering considerations in data center design. Data center equipment is extremely heavy — UPS battery banks, generators, and rack systems create substantial floor loading. Placing a data center on an upper floor would require a structural engineering review to verify that the floor can bear the load. In Ethiopia, seismic analysis is an additional consideration that is sometimes overlooked but should always be included in the design process, given the regional geological context. Flood risk assessment is also mentioned as a site consideration, particularly for ground-floor facilities.

4.4 Disaster Recovery Site Requirements

The session included a professional explanation of Disaster Recovery (DR) site design. The data center staff explained that while the AAiT 5-Kilo and AAU 6-Kilo campus data centers serve as mutual backup sites for each other, a true DR site must be separated from the primary site by a significant distance

— approximately 100 kilometers or more. This ensures that a single environmental or regional event (earthquake, fire, major power outage) cannot simultaneously affect both the primary and DR sites. The mutual proximity of the two AAU campus data centers means the current arrangement functions more as a warm standby than a fully compliant DR solution.

4.5 Historical Context — Previous Data Center Location

The staff shared that the current data center is not the first at AAIT. An earlier data center was located on the electrical floor of a different building on campus. As the university's digital service demands grew, that facility became insufficient — both in terms of physical space and its ability to scale. The decision was made to build a purpose-designed data center in a new location, with scalability, proper containment, and expansion capacity built in from the beginning. This historical context underscored the importance of forward planning in data center design.

5. Practical Experience and Knowledge Gained

The datacenter visit provided a uniquely valuable opportunity to observe how theoretical concepts taught in networking and systems courses are implemented in a real, operational environment. Several specific connections between academic content and practical reality were reinforced:

- **Network topology:** The collapsed core architecture observed in the AAiT data center is discussed in networking coursework as a cost-effective approach for medium-scale environments. Seeing it implemented physically — with actual fiber patch panels connecting to core switches, and those core switches routing traffic to Ethio Telecom and to campus buildings — made the concept tangible in a way that diagrams alone cannot achieve.
- **Single-mode vs. multi-mode fiber:** The explanation of fiber type selection based on distance gave practical context to what is often presented as a purely theoretical choice. Understanding that distances of 400–500 meters make single-mode fiber the appropriate choice connects the technical specification to real infrastructure decision-making.
- **Power architecture:** The layered power model (utility → AVR → ATS → UPS → IT load, backed by generators) is a standard topic in data center courses. Observing the actual equipment — the physical UPS units with their visual displays, the battery banks, the ATS panel — made the architecture real and understandable.
- **Heat management:** The hot aisle / cold aisle containment strategy is a classic data center cooling concept. Seeing the physical racks arranged to separate hot and cold airflows, with outdoor AC units on the building exterior handling heat rejection, connected the concept directly to engineering reality.
- **Monitoring integration:** The ECC 800 and NetEco architecture showed how a modern data center integrates physical sensors, power devices, and cooling systems into a single monitoring platform — a practical example of the kind of system integration discussed in systems administration and DCIM courses.
- **Redundancy design:** The consistent emphasis on redundancy at every level — generators, UPS, core switches, firewalls, fiber paths, campus links — reinforced the engineering principle that reliable systems are built by eliminating single points of failure at every layer.
- **Scalability planning:** The 30% current utilization of rack capacity, combined with the modular expansion capability of the FusionModule 2000, illustrated the importance of designing systems for future growth rather than only for current requirements.

6. Key Lessons and Insights

#	Lesson	Application
1	Design is always business-driven	No universal data center design exists. Redundancy levels, power architecture, and cooling capacity must match the criticality of services hosted.
2	Heat management is the primary operational challenge	Every watt consumed generates heat. Hot/cold aisle containment, proper AC sizing, and sensor-based monitoring are essential operational tools.
3	Redundancy must exist at every layer	Single points of failure at any layer — power, network, cooling, storage — can cause total outages. Dual systems are the minimum for critical services.
4	Fiber type selection is distance-dependent	Single-mode fiber is required for campus-scale distances (>400m). Multi-mode variants suit short-range intra-building links.
5	Collapsed core simplifies medium-scale networks	Eliminating the distribution layer reduces cost and management complexity while maintaining acceptable performance for institutional networks.
6	Ground floor placement is structurally critical	Data center equipment is extremely heavy. Ground floor placement avoids the need for expensive structural reinforcement.
7	Single-vendor integration reduces operational complexity	Mixing vendors for servers, cooling, and monitoring creates integration challenges and requires more specialized staff.
8	A true DR site requires geographic separation	Proximity-based backup cannot protect against regional disasters. 100+ km separation is the professional standard for DR site planning.
9	Scalability must be built in from day one	Designing to 30% current utilization with a modular expansion path allows the facility to grow without full rebuilds.
10	Staging new equipment protects production systems	Evaluating devices before connecting them to production infrastructure is a practical security measure against rogue or compromised hardware.

Table 7: Key lessons and their practical applications

7. Challenges, Risks, and Operational Considerations

Based on the observations and discussions during the visit, the following challenges and operational risks were identified at the AAiT data center:

7.1 Power Grid Reliability

Ethiopia's electrical grid experiences fluctuations and outages that are common in developing economies. The data center's multi-layered backup power system (AVR, UPS, dual generators) directly addresses this risk. However, an extended utility outage lasting longer than the battery runtime (~8 hours) without generator availability would represent a serious operational risk, and fuel supply logistics for the generators must be actively managed.

7.2 Heat Density Management

As server utilization increases and higher-density computing equipment is deployed, the thermal load on the cooling systems will increase proportionally. The current cassette AC and hot/cold aisle arrangement must be properly maintained and scaled in parallel with IT capacity growth. A cooling failure is one of the fastest paths to equipment damage in a data center environment.

7.3 Current Rack Underutilization

The data center currently operates at approximately 30% of its rack capacity. While this was a deliberate design decision to accommodate future growth, underutilization represents a form of investment inefficiency if growth does not materialize as forecasted. Active capacity planning and workload consolidation strategies should be periodically reviewed.

7.4 Disaster Recovery Site Limitations

The mutual DR arrangement between the 5-Kilo and 6-Kilo campuses, while providing data redundancy within the city, does not satisfy the geographic separation requirement for a formally compliant DR site. A significant citywide event could affect both sites simultaneously. Establishing a remote DR site (100+ km) is the long-term requirement for mission-critical services.

7.5 Seismic and Environmental Site Risk

Addis Ababa sits in the East African Rift Valley, a seismically active region. While the data center staff acknowledged that seismic analysis is not commonly practiced in Ethiopian data center design, it is a valid engineering consideration for a facility housing critical infrastructure. Additionally, ground-floor placement while generally structurally advantageous also presents a marginal flood risk that should be addressed through adequate drainage and raised floor design.

8. Personal Reflection

This data center visit was one of the most informative practical experiences of my academic studies at AAU CTBE. Having studied networking, systems architecture, and software engineering in classrooms for several years, I often found it difficult to truly connect the abstract concepts in textbooks with the physical reality of how digital services are actually delivered and sustained.

Walking through the power room and seeing the UPS units, battery banks, and ATS panels physically installed — and listening to the data center engineer explain precisely how power is routed through each device and how the system automatically responds to a power failure — made the entire topic of power infrastructure completely concrete for me. Similarly, seeing the fiber patch panels that connect every building on the campus, and understanding that our daily use of AAU's student portal and LMS platforms depends on those physical fibers running through the campus, gave me a new appreciation for the engineering effort that underlies what users take for granted.

The professional insights shared by the data center staff — particularly on topics like business-driven design, DR site requirements, and the historical evolution of the facility from its original location on the electrical floor to the current purpose-built data center — gave me a perspective on data center engineering that goes well beyond what any textbook covers. These are the kinds of practical decision-making frameworks that engineers develop through experience, and having them shared openly during the visit was genuinely valuable.

I left the visit with a stronger motivation to pursue deeper study in network infrastructure, systems administration, and data center engineering — fields I now see not as abstract technical disciplines but as the concrete foundation on which digital services for millions of people are built and maintained.

9. Recommendations

Based on the observations made during the visit and the discussions with data center personnel, the following recommendations are offered for the continued development and improvement of the AAiT data center:

- **Establish a formal remote DR site:** The current mutual backup arrangement with the 6-Kilo campus does not provide the geographic separation required for a formally compliant disaster recovery configuration. A remote DR site at a distance of at least 100 km should be established as a long-term strategic objective for mission-critical services.
- **Pursue incremental utilization growth:** With racks currently at approximately 30% utilization, there is significant available capacity. A structured capacity management plan that progressively onboards additional workloads — including AAU's Moodle, e-learning, and research platforms — would improve operational cost-efficiency.
- **Conduct periodic seismic risk assessment:** Given Addis Ababa's location in the East African Rift Valley, a formal seismic risk assessment for the data center should be incorporated into periodic infrastructure reviews, particularly before major hardware expansion projects.
- **Expand student practical access:** The data center's value as a teaching facility could be enhanced by creating structured lab sessions that allow networking and software engineering students to gain hands-on experience with the equipment under supervised conditions — beyond the current tour-only format.
- **Consider formal TIA-942 alignment review:** While the facility is designed with reference to TIA-942 principles, a formal internal alignment review against TIA-942 Rated-2 criteria would identify any gaps between current implementation and the standard, and provide a roadmap for certification-readiness.
- **Implement automated fuel management for generators:** Proactive monitoring of generator fuel levels and automated alerts should be integrated into the NetEco monitoring system to ensure generator readiness is maintained without reliance on manual inspection schedules.

10. Conclusion

The visit to the AAiT Data Center at the 5-Kilo campus of Addis Ababa University provided a comprehensive and deeply informative practical experience that significantly enhanced my understanding of real-world data center infrastructure, design principles, and operational practices.

The facility exemplifies a well-planned academic data center that has evolved from modest origins to a modern, modular infrastructure capable of supporting the digital needs of one of Africa's largest universities. Its multi-layered power architecture, fiber-optic networking backbone, Huawei FusionModule 2000 containment system, FM-200 fire suppression, centralized ECC 800 / NetEco monitoring, and hot/cold aisle cooling represent a coherent and professionally implemented data center design.

The experience-sharing session conducted by the data center staff was equally valuable — providing professional context on business-driven design decisions, the practical tradeoffs between different technologies and architectures, and the real-world challenges of managing critical infrastructure in an Ethiopian university context. The discussion of topics such as DR site design, seismic considerations, fiber type selection, and modular scalability gave practical depth to concepts that are often presented abstractly in academic settings.

As a software engineering student, this visit reinforced that software systems do not exist in isolation — they depend entirely on physical infrastructure that must be carefully designed, reliably powered, actively cooled, securely protected, and continuously monitored. Understanding this physical foundation is essential knowledge for any engineer who designs, deploys, or manages digital systems at scale.

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